Community Pharmacy Inventory and Sales Management System

Report 1: Project Description Report

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1. Business Idea and Background

The proposed system is a small information system for a community pharmacy. It supports the recording of products, product categories, suppliers, purchase orders, sales, and the staff members who record transactions. It is a design proposal for a UML modelling assignment; it is not a claim that a working application has been implemented.

2. Problem Statement

A pharmacy may rely on paper notes or disconnected spreadsheets to track stock, supplier orders, and sales. Such practices can make it difficult to identify low-stock products, calculate transaction totals consistently, and trace which employee recorded a transaction. The proposed model describes a structured way to represent these business records and their relationships.

3. Purpose and Objectives

The purpose is to model the main data concepts and business relationships needed for basic pharmacy inventory and sales management. The objectives are to represent products and categories; record supplier purchase orders and their line items; record sales and sale line items; associate transactions with employees; optionally associate a sale with a customer; and define clear multiplicities and class responsibilities.

4. Project Scope

Included: product details and stock quantity; category assignment; supplier contact details; purchase orders and purchase-order lines; sales and sale lines; transaction totals; employee attribution; optional customer details; low-stock checking; and basic order/sale status.

Excluded: prescription validation, clinical advice, insurance claims, tax and accounting integration, barcode scanning, online ordering, delivery logistics, multi-branch synchronisation, authentication/security implementation, supplier payment processing, and a deployed software application. These are outside the scope of this first UML model.

5. Stakeholders

The primary stakeholders are pharmacy employees, the pharmacy manager or owner, suppliers, and customers. Employees record sales and purchase orders; the manager reviews stock and transactions; suppliers provide products; customers may be recorded against a sale when their details are needed.

6. Functional Requirements

FR1: The system shall represent each product with an identifier, name, unit price, stock quantity, and reorder level.

FR2: The system shall group products into categories; each product belongs to exactly one category in this model.

FR3: The system shall store supplier contact information and represent purchase orders placed with a supplier.

FR4: Each purchase order shall contain one or more purchase-order lines, each referring to one product and recording quantity and unit cost.

FR5: The system shall record sales with date/time, payment method, status, and one or more sale lines.

FR6: Each sale line shall refer to one product and record quantity and the unit price charged for that sale.

FR7: Each sale and purchase order shall be attributed to one employee; an employee may record zero or many transactions of either kind.

FR8: A sale may optionally be associated with one customer; a customer may be associated with zero or many sales.

FR9: The model shall support calculating line totals and transaction totals and checking whether a product is at or below its reorder level.

7. Non-Functional Requirements

The model should be understandable to first-year Computer Science students, use consistent names and data types, avoid duplicated transaction-line data, and use standard UML notation. A future implementation should protect customer and transaction data, validate non-negative quantities and prices, and maintain accurate stock records; these are design considerations rather than implemented features.

8. Main Classes and Responsibilities

Product: Represents a medicine or other stocked pharmacy item.

Category: Groups products by a pharmacy category.

Supplier: Stores details of a business that supplies pharmacy products.

PurchaseOrder: Records an order placed with a supplier.

PurchaseOrderLine: Records one product and quantity within a purchase order.

Sale: Records a customer transaction processed by an employee.

SaleLine: Records one product, quantity and sale price within a sale.

Customer: Stores optional customer details for a sale.

Employee: Represents a staff member who records and manages transactions.

9. Assumptions and Constraints

A product belongs to exactly one category. A purchase order and a sale each contain at least one line. Each order and sale is recorded by exactly one employee. A sale can be recorded without identifying a customer. Each line refers to exactly one product. PurchaseOrderLine and SaleLine are composition parts: in this model, a line has no independent business existence outside its owning transaction. Product stock is represented as a single integer quantity; batch numbers, expiry dates, and multiple storage locations are not modelled. Decimal is used for money to avoid implying binary floating-point arithmetic.

10. Conclusion

This project defines a manageable pharmacy business domain with inventory, purchasing, and sales concepts. Its class model makes the key records and constraints explicit and provides a consistent basis for discussion or future implementation. The deliverable is a UML design, not a completed application.